

Lecture 6: Statistical Inference

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6.1 Where We Are in the Course

The first five lectures built the modeling side of the course:

- Lecture 1 introduced simple linear regression.
- Lecture 2 extended that framework to multivariable linear regression and design matrices.
- Lecture 3 moved to logistic regression for binary outcomes.
- Lecture 4 organized those ideas into GLMs and GAMs, emphasizing that the response model and the mean-function flexibility both matter.
- Lecture 5 made the warning explicit: models do what they are told to do, so specification choices determine which patterns the model can and cannot discover.

So far, the emphasis has been on writing down models, fitting them, and interpreting what they say. This lecture adds the next layer: once a pattern appears in the output, how do we judge whether it is stronger than what chance variation alone could plausibly produce?

Inference does *not* rescue a bad model. If the model is misspecified or the comparison is confounded, then a small p-value can still be answering the wrong question. What inference adds is a disciplined way to quantify how surprising the observed pattern would be under a stated null explanation.

For example:

- A basketball team shoots better after three days of rest. Is the rest effect real, or is this just game-to-game noise?
- A baseball player's batting average improves after a coaching change. Is this evidence that the coach helped, or is this regression to the mean?
- A diving judge gives higher scores to divers from his own country. Is that evidence of nationality bias, or could that happen by chance?
- A coefficient in a logistic regression model is positive. Is it large compared to the uncertainty in the estimate?

The key conceptual tool will be the *null distribution*. A test statistic becomes evidence only after we know what values that statistic would usually take if the “boring” or “simple” explanation were true.

Main idea. Statistical significance is not about whether an observed effect is large in isolation. It is about whether the observed effect is large relative to a null distribution.

6.2 Chance Variation

Sports data are noisy because athletes, teams, judges, and opponents vary in ways that are not fully explained by our measured variables. This unexplained variation is not always a mistake. It is often an unavoidable feature of the data-generating process.

Definition 6.1 (Chance Variation). *Chance variation is the random-looking variation in observed data that remains even when there is no systematic effect of interest.*

A player can hit a hard line drive directly at a fielder. A great shooter can miss an open three. A judge can score one dive slightly higher or lower than another judge simply because scoring is subjective. These events do not automatically imply that our model is wrong. They mean that data contain irreducible variability.

The question is not whether chance variation exists. It does. The question is:

How large would an observed pattern need to be before chance variation no longer feels like a reasonable explanation?

This question is the starting point for significance testing.

6.3 The Four Ingredients of a Statistical Test

A statistical test requires four ingredients.

1. A **null hypothesis**, which formalizes the simple explanation.
2. A **test statistic**, which turns the data into a number measuring evidence against the null.
3. A **null distribution**, which says what values of the test statistic are expected under the null.
4. A **p-value**, which compares the observed test statistic to the null distribution.

Definition 6.2 (Null Hypothesis). *The null hypothesis, denoted H_0 , is the hypothesis we temporarily assume to be true in order to measure how surprising the observed data are under a simple baseline explanation.*

Definition 6.3 (Research Hypothesis). *The research hypothesis, often denoted H_A or H_1 , is the claim we are looking for evidence in favor of. It is usually the scientific or sports question that motivated the analysis.*

The null is not always the hypothesis we believe. It is the hypothesis that gives us a benchmark for chance variation.

Definition 6.4 (Test Statistic). *A test statistic is a function of the data,*

$$T = T(\text{data}),$$

chosen so that larger, smaller, or more extreme values represent stronger evidence against the null hypothesis.

For example, if we are studying whether a judge favors divers from his own country, a natural statistic is the difference between the judge's average discrepancy for same-country divers and his average discrepancy for other divers.

Definition 6.5 (Null Distribution). *The null distribution is the distribution of the test statistic T when the null hypothesis is true.*

The null distribution is the central object in this lecture. Without it, we cannot say whether an observed statistic is surprising.

6.4 Exact Null Distributions: Coin Flipping

Sometimes the null distribution is exact and easy to write down.

Example 6.6 (A Fair Coin). Suppose we flip a coin n times and observe X heads. We want to test whether the coin is biased toward heads.

The null hypothesis is

$$H_0 : p = \frac{1}{2},$$

where p is the probability of heads.

A natural test statistic is

$$T = X = \text{number of heads.}$$

Under the null hypothesis, every flip is a Bernoulli random variable with probability $1/2$ of heads, so

$$T \sim \text{Binomial}\left(n, \frac{1}{2}\right).$$

Therefore the null distribution is known exactly:

$$\mathbb{P}_{H_0}(T = k) = \binom{n}{k} \left(\frac{1}{2}\right)^n, \quad k = 0, 1, \dots, n.$$

If we flip the coin 20 times and observe 16 heads, then the one-sided p-value for bias toward heads is

$$\begin{aligned} p\text{-value} &= \mathbb{P}_{H_0}(T \geq 16) \\ &= \sum_{k=16}^{20} \binom{20}{k} \left(\frac{1}{2}\right)^{20} \\ &\approx 0.0059. \end{aligned}$$

This calculation is possible because the null distribution is explicit. We know what chance variation looks like if the coin is fair.

6.4.1 One-sided and Two-sided Alternatives

If the research hypothesis is that the coin is biased *toward heads*, then large values of T are evidence against the null. The p-value is

$$\mathbb{P}_{H_0}(T \geq t_{\text{obs}}).$$

If the research hypothesis is that the coin is biased in *either direction*, then both unusually large and unusually small values of T are evidence against the null. A natural two-sided p-value is

$$\mathbb{P}_{H_0} \left(\left| T - \frac{n}{2} \right| \geq \left| t_{\text{obs}} - \frac{n}{2} \right| \right).$$

For $n = 20$ and $t_{\text{obs}} = 16$, the two-sided p-value is approximately 0.0118.

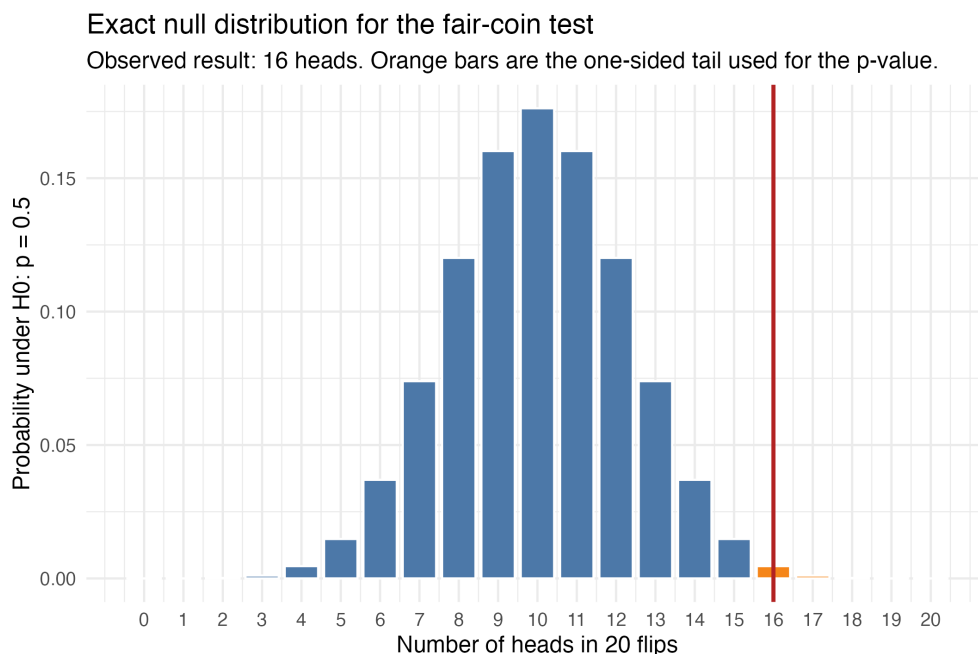


Figure 6.1: Exact null distribution for the number of heads in 20 flips of a fair coin. The red line marks the observed result, 16 heads, and the orange bars show the one-sided tail used for the p-value.

6.5 p-values

Definition 6.7 (p-value). *A p-value is the probability, assuming the null hypothesis is true, of obtaining a test statistic at least as extreme as the one observed.*

For a one-sided test where larger values of T are more extreme,

$$p\text{-value} = \mathbb{P}_{H_0}(T \geq t_{\text{obs}}). \quad (6.1)$$

For a two-sided test where deviations in either direction are important,

$$p\text{-value} = \mathbb{P}_{H_0}(|T| \geq |t_{\text{obs}}|), \quad (6.2)$$

provided $T = 0$ represents the null value. The exact two-sided definition can depend on the statistic and the shape of the null distribution, but the idea is always the same: compare the observed result to what the null says is typical.

Interpretation. A small p-value means that the observed statistic would be unusual if the null hypothesis were true. It does not, by itself, prove that the research hypothesis is true.

Common misinterpretations:

- The p-value is not the probability that the null hypothesis is true.
- The p-value is not the probability that the result happened by chance.
- The p-value is not the size of the effect.
- A statistically significant result is not automatically a practically important result.
- A statistically significant association is not automatically a causal effect.

6.6 Statistical Significance

Definition 6.8 (Statistical Significance). *An observed result is statistically significant at level α if its p-value is at most α .*

For example, at the $\alpha = 0.05$ level, a result is statistically significant if

$$p\text{-value} \leq 0.05.$$

The number 0.05 is a convention, not a law of nature. It is a threshold for deciding when the observed statistic is sufficiently unusual under the null. In sports analytics, it is often better to report the effect size, uncertainty interval, and p-value rather than reducing the analysis to “significant” or “not significant.”

6.7 A Model-Based Null Distribution: Normality as an Assumption

The coin-flipping example is clean because the null distribution is exact. In many sports problems, we do not have such a simple exact null distribution.

Suppose $Y_1, \dots, Y_n$ are measurements of a player’s performance under some condition. For example, Y_i could be a shooting metric across games, a running speed measurement, or a judge’s scoring discrepancy. Suppose we want to test whether the mean is equal to some null value μ_0 .

A model-based approach might assume

$$Y_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2).$$

Under the null hypothesis $H_0 : \mu = \mu_0$, if σ is known, the statistic

$$Z = \frac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}}$$

has the standard normal null distribution,

$$Z \sim \mathcal{N}(0, 1).$$

This gives a p-value. For a two-sided test,

$$p\text{-value} = \mathbb{P}_{H_0}(|Z| \geq |z_{\text{obs}}|).$$

Remark 6.9 (Normality in this lecture). We have not yet derived the Central Limit Theorem. Therefore, in this lecture, normality should be understood as a modeling assumption or approximation, not as something we have proved from first principles. Later, the Central Limit Theorem will explain why normal approximations often appear even when the raw data are not normal.

This is an important philosophical point. A null distribution is always produced by assumptions. Sometimes those assumptions are nearly built into the problem, as in fair coin flipping. Sometimes they are modeling assumptions, as in Gaussian errors. Sometimes they come from a randomization design or an exchangeability assumption, as in permutation testing.

6.8 Regression Output and Null Distributions

In previous lectures, we fit regression-type models. Those models also contain null distributions, even if software hides the construction.

A regression p-value is not magic. It comes from a statement like this:

If the null hypothesis $H_0 : \beta_j = 0$ were true, what values of the estimated coefficient $\hat{\beta}_j$ would we expect to see just from chance variation?

6.8.1 Two Layers of a Regression Model

Regression models usually make two related claims.

1. A claim about the **outcome distribution** conditional on predictors.
2. A claim about the **sampling distribution** of coefficient estimates or test statistics.

For example, in linear regression we often write

$$Y_i \mid X_i = x_i \sim \mathcal{N}(x_i^T \beta, \sigma^2).$$

This is a distributional assumption about the outcome Y_i given predictors X_i .

From this assumption, and from the geometry of least squares, we get a distribution for coefficient estimates. In the classical Gaussian linear model,

$$\hat{\beta}_j \approx \mathcal{N}(\beta_j, \text{se}(\hat{\beta}_j)^2),$$

and, under $H_0 : \beta_j = 0$,

$$\frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)}$$

has a known or approximately known null distribution.

Remark 6.10 (Parameters versus estimators). In frequentist statistics, the coefficient β_j is treated as a fixed but unknown number. The random object is $\hat{\beta}_j$, the estimate computed from data. When people informally say “the beta is normal,” they usually mean that $\hat{\beta}_j$ has an approximately normal sampling distribution around β_j . In Bayesian statistics, one may instead put a probability distribution directly on β_j . In this lecture, we use the frequentist convention.

6.8.2 Examples Across Models

Model	Outcome assumption	Mean model	Coefficient inference
Linear regression	$Y_i x_i \sim \mathcal{N}(\mu_i, \sigma^2)$	$\mu_i = x_i^T \beta$	$\hat{\beta}_j$ normal/t
Logistic regression	$Y_i x_i \sim \text{Bernoulli}(p_i)$	$\text{logit}(p_i) = x_i^T \beta$	$\hat{\beta}_j$ approx. normal
Poisson regression	$Y_i x_i \sim \text{Poisson}(\lambda_i)$	$\log(\lambda_i) = x_i^T \beta$	$\hat{\beta}_j$ approx. normal
GAMs	Depends on outcome	$g(\mu_i) = \eta_i$ with smooth terms	Approximate tests for terms

Table 6.1: Regression models specify a distribution for the outcome and imply a null distribution for coefficient estimates or test statistics.

For logistic and Poisson regression, the normality of $\hat{\beta}$ is typically an approximation justified by large-sample likelihood theory. Since we have not developed that theory yet, the right interpretation for now is:

Software p-values in GLMs rely on model-based approximations to the null distribution of a coefficient estimate or related statistic.

This connects directly to model misspecification. If the outcome distribution, independence assumptions, variance assumptions, or functional form are badly wrong, then the null distribution used by the software may be wrong too. In that case, the p-value may no longer mean what it claims to mean.

6.9 Permutation Tests: Building a Null Distribution from the Data

Now consider a setting where a clean parametric null distribution is hard to justify. This is common in sports analytics. Diving scores, referee calls, subjective ratings, and play-by-play outcomes often contain complicated dependence and non-normality.

Permutation testing gives us another way to construct a null distribution.

6.9.1 The Core Idea

A permutation test is appropriate when the null hypothesis implies that certain labels are exchangeable.

Definition 6.11 (Exchangeability for a Permutation Test). *Labels are exchangeable under the null if, assuming the null hypothesis is true, the observed outcomes would have been equally plausible under many rearrangements of the labels.*

For example, suppose we have n observations, each with an outcome Y_i and a group label $G_i \in \{0, 1\}$. We want to test whether the group labels are associated with the outcomes.

A natural statistic is the difference in sample means:

$$T_{\text{obs}} = \bar{Y}_{G=1} - \bar{Y}_{G=0}. \quad (6.3)$$

Under the null hypothesis of no relationship between G and Y , the labels G_i can be permuted among the outcomes Y_i . Each permutation gives a new test statistic:

$$T^{(b)} = \bar{Y}_{G=1}^{(b)} - \bar{Y}_{G=0}^{(b)}, \quad b = 1, \dots, B.$$

The empirical distribution of $T^{(1)}, \dots, T^{(B)}$ is the permutation null distribution.

For a one-sided alternative where large values are evidence against the null, the Monte Carlo permutation p-value is often computed as

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbb{1}\{T^{(b)} \geq T_{\text{obs}}\}}{B + 1}. \quad (6.4)$$

The +1 correction prevents reporting a p-value of exactly zero when only finitely many random permutations are used.

6.9.2 Permutation Tests as a Response to the Null Distribution Problem

The permutation test does not remove assumptions. It changes the assumptions.

Instead of assuming a normal distribution for the outcomes, the permutation test assumes that the relevant labels are exchangeable under the null. This is often easier to believe in randomized experiments. In observational sports data, it still needs scrutiny.

Permutation logic. If the null hypothesis says the labels do not matter, then randomly rearranging the labels should produce datasets that are plausible under the null.

6.10 Case Study: Nationality Bias in Diving Judges

We now apply these ideas to a diving example. The research question is whether judges systematically give higher scores to divers from their own country.

6.10.1 Discrepancies

For each judged dive, define a judge's discrepancy as

$$D_i = \text{judge's score on dive } i - \text{baseline score for dive } i.$$

The baseline could be the average score for that dive across judges or another measure of the dive's quality. The purpose of using discrepancies is to compare judges after accounting for the fact that some dives are simply better than others.

The key idea is that raw scores are not enough. If American divers in a judge's sample happened, on average, to perform stronger dives or execute them better than non-American divers, then that judge's raw average score for Americans could be higher even without nationality bias. Discrepancies try to remove overall dive quality first, then ask whether the judge tends to score *above baseline* more often for nationality-matched divers.

For a given judge, define

$$M_i = \begin{cases} 1, & \text{if the diver's nationality matches the judge's nationality,} \\ 0, & \text{otherwise.} \end{cases}$$

The test statistic is the difference of discrepancies:

$$\text{DoD} = \bar{D}_{M=1} - \bar{D}_{M=0}. \quad (6.5)$$

A positive DoD means that the judge gives higher discrepancies, on average, to divers from the judge's own country than to other divers.

6.10.2 Why This Statistic Matches the Question

For each dive in a judge's sample, we now have two pieces of information:

- a discrepancy D_i , which says whether the judge scored that dive above or below baseline;
- a match indicator M_i , which says whether the diver shared the judge's nationality.

The DoD statistic asks a clean question:

Among the dives this judge saw, are the nationality-matched dives disproportionately the ones with positive discrepancies?

That is much closer to the substantive question of nationality bias than simply comparing raw average scores across countries.

6.10.3 Hypotheses

The research hypothesis is:

H_A : the judge is biased in favor of divers from the judge's own country.

The null hypothesis is:

H_0 : the judge's discrepancies are not systematically related to nationality matches.

Under the null, after we have computed discrepancies, the nationality-match labels should be exchangeable with respect to the discrepancies. That is, the set of discrepancy values would be just as plausible if the "matched nationality" labels were attached to a different set of dives.

Equivalently, once overall dive quality has been adjusted for through the discrepancy, the matched dives should not systematically line up with unusually high discrepancy values if there is no nationality effect.

6.10.4 McFarland Example

For Judge McFarland, the observed data can be summarized as follows:

- He judged 657 dives.
- 42 of those dives involved American divers, matching his nationality.
- His average discrepancy for matched dives was about 0.20.
- His average discrepancy for non-matched dives was about 0.01.
- The observed DoD was about 0.19.

The observed difference is positive. But that alone is not enough. We need to ask whether a DoD of 0.19 is large relative to the null distribution.

One useful way to read these numbers is:

After adjusting for dive quality through the baseline, McFarland scored American divers about 0.19 points higher, on average, than other divers.

The inferential question is whether that remaining gap is too large to attribute to random assignment of the 42 matched-nationality dives among the 657 dives he judged.

6.10.5 Permutation Procedure

For Judge McFarland, the permutation test proceeds as follows.

1. Keep McFarland's 657 discrepancy values fixed.
2. Randomly choose 42 of the 657 dives to be labeled "matched." Label the remaining 615 dives "non-matched."
3. Compute the DoD for this randomized labeling.
4. Repeat many times to form a permutation null distribution.
5. Compare the observed DoD, 0.19, to this permutation null distribution.

Step 2 is the heart of the procedure. It keeps fixed the number of nationality-matched dives that McFarland actually saw, but it breaks any systematic connection between nationality and scoring. Under the null, this reshuffling should generate the kind of DoD values we would expect from chance alone.

In the observed analysis, only about 1 out of 1000 permuted assignments produced a DoD as large as McFarland's actual value. Equivalently, the permutation p-value was around 0.001.

This means that if nationality-match labels were irrelevant to McFarland's discrepancies, a DoD as large as the one observed would be very unusual under the permutation null distribution.

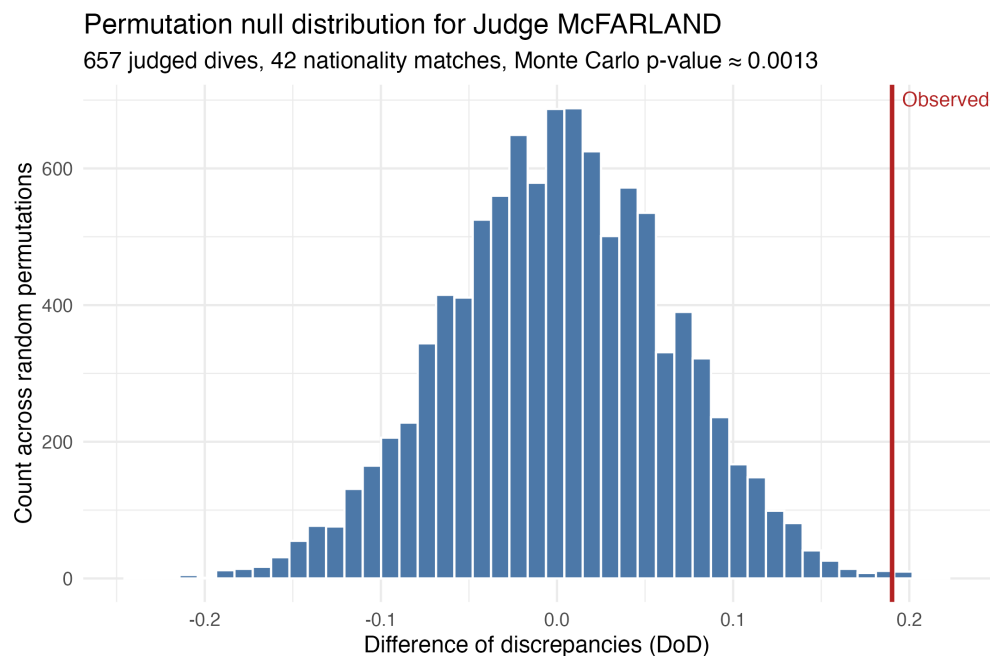


Figure 6.2: Permutation null distribution for McFarland's difference of discrepancies. The observed DoD is far out in the right tail, which is why the permutation p-value is very small.

This is statistical evidence against the null hypothesis of no nationality-related discrepancy. It is not, by itself, a complete explanation of why the pattern exists. It also depends on the validity of the discrepancy definition and the exchangeability assumption.

6.10.6 A Small Sample Can Still Be Informative

Small samples are not automatically useless. They are weak when the observed data are not extreme. But very extreme data can be informative even when the sample size is small.

Suppose a judge has only 3 nationality-matched dives, but all three discrepancies are in the top 10% of that judge's discrepancy distribution. A rough approximation to the p-value is

$$(0.10)^3 = 0.001.$$

The exact permutation calculation would sample without replacement, so it is not exactly the same as multiplying independent probabilities. But the approximation explains the intuition: seeing all three matched dives among the most favorable discrepancies would be unusual under random labeling.

6.11 What Assumptions Buy Us

The theme of this lecture is that inference requires a null distribution. Different methods get that null distribution in different ways.

Method	Source of null distribution	Main assumption
Coin-flip test	Exact binomial distribution	Independent fair flips under H_0
Normal mean test	Normal model	Gaussian outcomes/errors
Regression coefficient test	Regression/GLM model	Correct outcome and error model, or good approximation
Permutation test	Re-labeling the data	Exchangeability under H_0

Table 6.2: Every test needs a null distribution, and every null distribution comes from assumptions.

Assumptions are not automatically bad. They are what allow us to learn from incomplete information. But we need to know what assumptions we are making.

A useful statistical habit is to ask:

1. What is the test statistic?
2. What is the null hypothesis?
3. What is the null distribution?
4. What assumptions produce that null distribution?
5. Are those assumptions plausible in this sports context?

6.12 Significance versus Practical Importance

Statistical significance answers one question:

Is the observed statistic surprising under the null distribution?

It does not answer:

Is the effect large enough to matter for decision-making?

In sports analytics, this distinction is critical. A tiny effect can be statistically significant in a massive dataset. A large effect can fail to be statistically significant in a small dataset. Neither statement alone tells a coach, front office, or player what to do.

For any analysis, report both:

- an **effect size**, such as a difference in points, win probability, goals, expected points, or log-odds;
- an **uncertainty measure**, such as a standard error, confidence interval, or p-value.

6.13 Multiple Testing

The diving example naturally leads to a multiple-testing problem. If we test many judges, some small p-values can occur by chance even if every null hypothesis is true.

If we run 100 valid tests at the 0.05 level, and all null hypotheses are true, then the expected number of statistically significant results is

$$100 \times 0.05 = 5.$$

So if we screen many judges, a few small p-values are not surprising by themselves. Two common error targets are:

- **FWER (family-wise error rate):**

$$\text{FWER} = \mathbb{P}(V \geq 1),$$

the probability of making *at least one* false discovery in the whole family of tests.

- **FDR (false discovery rate):**

$$\text{FDR} = \mathbb{E} \left[\frac{V}{\max(R, 1)} \right],$$

the expected fraction of reported discoveries that are false.

Here V is the number of false discoveries and R is the total number of rejections.

These targets correspond to different goals:

- If falsely accusing even one judge would be a major problem, controlling **FWER** makes sense. The standard simple tool is **Bonferroni**, which is conservative but easy to explain.
- If the goal is exploratory screening, where we are willing to tolerate some false leads in exchange for finding more real ones, controlling **FDR** is often more appropriate. The standard tool is **Benjamini–Hochberg**.

Therefore, when many tests are performed, we should be cautious. In practice:

- pre-specifying the main test before looking at the data;
- adjusting for multiple comparisons, using a target such as FWER or FDR;
- reporting all tested hypotheses, not only the significant ones;
- treating surprising results as exploratory unless confirmed on new data.

This is another place where statistical thinking matters. The p-value for one judge answers a conditional question about that judge and that statistic. It does not automatically account for the fact that we may have searched across many judges, many sports, or many outcomes.

6.14 Significance Is Not Causality

This lecture is a bridge to causal inference. Statistical significance tells us whether an observed association is surprising under a null model. It does not tell us whether the association is causal.

Another way to say this is:

Significance can rule out one particular chance-based null explanation. It does not rule out confounding, selection, reverse causality, or model misspecification.

For example, suppose a regression model finds that NBA players score more points in games where they play more minutes, and the coefficient is highly statistically significant. That does not mean that simply giving any player more minutes would cause that player to score at the same rate. Better players are selected to play more minutes.

Similarly, suppose a baseball model finds that players who changed coaches improved more than players who did not. That may be a coaching effect, but it may also reflect selection, regression to the mean, injury recovery, or other confounding variables.

The next major conceptual step is causal inference, where we ask:

When can an association be interpreted as a cause-and-effect relationship?

Significance is still useful there, but it is not sufficient. Causal inference requires both:

1. **Identification:** are we estimating the causal quantity we care about?
2. **Inference:** how uncertain is our estimate?

This lecture focused on inference. Causal inference will add identification.

6.15 Summary

The main lessons are:

- Chance variation is unavoidable in sports data.
- A test statistic is only interpretable after we know its null distribution.
- A p-value is the probability, under the null, of a result at least as extreme as the one observed.
- Exact null distributions exist in simple cases like coin flipping.
- Model-based null distributions rely on assumptions such as normality, Bernoulli outcomes, Poisson outcomes, independence, and correct functional form.
- Regression p-values come from null distributions for coefficient estimates or test statistics.
- Permutation tests build a null distribution by re-labeling data under an exchangeability assumption.
- Statistical significance is not the same as practical importance.
- Statistical significance is not the same as causality.

6.16 Exercises

1. A coin is flipped 30 times and lands heads 22 times. Write the null hypothesis, test statistic, null distribution, and one-sided p-value for testing whether the coin is biased toward heads.
2. In a basketball dataset, a team wins 18 of 25 games after two or more days of rest. Suppose the team's baseline win probability is 0.50. Under the null model that rest has no effect and each game is independent, what is the null distribution for the number of wins? What is the one-sided p-value?

3. In the diving example, explain why comparing a judge's raw average score for American divers to his raw average score for non-American divers is not enough to establish nationality bias.
4. Define the difference of discrepancies statistic for a judge. Why is it a better statistic than raw average score difference?
5. In your own words, explain the exchangeability assumption in the permutation test for diving judges.
6. Suppose a regression coefficient has estimate $\hat{\beta} = 0.12$ and standard error 0.04. Using a standard normal approximation, compute the two-sided p-value for testing $H_0 : \beta = 0$. What does this p-value mean?
7. Give an example of a statistically significant sports result that might not be practically important.
8. Give an example of a statistically significant association in sports that should not automatically be interpreted causally.